



A decomposition ensemble based deep learning approach for crude oil price forecasting

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ABSTRACT

As the price of crude oil has nonlinearity, instability, and randomness, capturing its behavior precisely is significantly challenging and leads to difficulties in forecasting. This study combines a decomposition-ensemble approach, optimized by the seagull algorithm, with a sentiment analysis to handle the problem. First, the cumulative sentiment score sequence is obtained by a sentiment analysis of news headlines data. Second, an adaptive signal decomposition method, namely, ensemble empirical mode decomposition (EEMD), is employed to decompose the historical crude oil future prices data into several intrinsic mode functions and one residual component to reduce the impact of noise. Third, a seagull optimization algorithm (SOA) is introduced to tune the hyperparameters of gated recurrent units (GRUs). The optimized GRU model is established to acquire the predicting values of each component integrated with the cumulative sentiment score sequence. Subsequently, multiple linear regression (MLR) is then introduced as the ensemble approach that integrates the forecasting results of each component. The empirical forecasting results of daily West Texas Intermediate (WTI) crude oil future and news headlines data validate our proposed decomposition-ensemble approach with different forecasting horizons. This approach significantly outperforms some other comparison models by means of forecasting accuracy and hypothesis tests. In addition, the EEMD components of WTI crude oil future price are reconstructed to analyze the impact of black-swan events on crude oil prices fluctuations from January 4, 2010 to September 17, 2019.

1. Introduction

As the most important industrial raw materials and strategic reserves, products refined from crude oil exert a great influence on social and economic development, and crude oil is an indispensable resource in the world. However, the rise of crude oil prices will push inflation, and its fluctuation has an important role in politics and economies worldwide, both of which are closely related to people's livelihoods and the development of countries. Thus, the accurate forecasting of oil futures prices can provide useful guidelines for both policy-makers and investors. However, the nonlinearity, instability, and randomness of crude oil futures prices are affected by various factors, such as the supply and demand relationship, inventories, and international events (Li et al., 2019a,b; Sun et al., 2018), rendering oil price forecasting an extremely difficult endeavor given the complex structure of oil prices (see Figs. 6 and 7).

In recent years, researchers have exploited a series of crude oil futures price prediction models to enhance forecasting accuracy. Based on the literature, predicting models comprise three categories: econometric models, machine learning approaches, and decomposition-ensemble methods. The classical econometric models, including the autoregressive integrated moving average (ARIMA) (Alquist & Kilian, 2010; Li et al., 2020), heterogeneous autoregressive type (HAR-type) (Sévi, 2014; Yang et al., 2018; Liu et al., 2020; Niu et al., 2021; Zhang et al., 2019) and generalized autoregressive conditional heteroscedasticity type (GARCH-type) (Cheong C. W., 2009; Wang et al., 2016; Klein and Walther, 2016; Lin et al., 2020; Zhang et al., 2018), are widely utilized in forecasting crude oil futures prices because their coefficients can be well explained. For example, Li et al. (2020) combined the ARIMA model with a deep learning model to predict financial time series to solve the deviation of the traditional ARIMA in forecasting high-frequency time series. The hybrid model integrates the superiorities of different models

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in characterizing the linear and nonlinear relationship of the error terms. Lin et al. (2020) applied several loss functions and evaluating indicators to prove that the proposed model, namely, GARCH-type models with hidden Markov (HM-EGARCH), outperforms the compared models. Based on linear, stationary, and regular hypotheses, these econometric models achieve suitable performance in forecasting accuracy. However, because of the nonlinear and nonstationary patterns in crude oil futures prices, such econometric models may fail to accurately model futures prices that have nonlinear, complex, and nonstationary characteristics.

Following the development of scientific and computational techniques, machine learning models, including the extreme learning machine (ELM) (Wang et al., 2018, 2021) and support vector machine (SVM) (Wang et al., 2020), are the most typical and are commonly utilized to predict crude oil prices. Furthermore, deep learning is becoming mainstream in machine learning because traditional machine learning approaches are shallow architectures. Recently, deep learning models, including artificial neural networks (ANNs) (Mirmirani and Li, 2004; Khoshalan et al., 2021), long short-term memory (LSTM) networks (Liu et al., 2022; Borovkova & Tsiamas, 2019; Urolagin et al., 2021), gated recurrent unit networks (GRUs) (Wang & Wang, 2020; Li and Wang, 2020), and convolutional neural networks (CNNs) (Li et al., 2019a,b), have been widely utilized in the domain of energy prediction. For example, Wang et al. (2018) constructed the Internet Concern index and combined it with the crude oil futures price to forecast futures price volatility through bivariate empirical mode decomposition (BEMD) and an ELM. Li et al. (2019a,b) made a simple attempt to apply a CNN based on online media text mining to crude oil price forecasting. Urolagin et al. (2021) exploited a new hybrid deep learning model, namely, a combined architecture of multivariate LSTM with Z score transformation and Mahalanobis distance, to predict crude oil prices. Wang and Wang (2020) applied a novel hybrid model, namely, a GRU with stochastic time effective weights (SW-GRU), to predict the components of energy prices decomposed by empirical mode decomposition (EMD). The proposed model achieved the best performance among all the selected models. Similarly, Li and Wang (2020) proposed a GRU with a stochastic time-intensity function (ST-GRU) to develop a new forecasting model to increase the prediction accuracy of the crude oil futures price. This method can make data more suitable for forecasting and can achieve excellent performances through empirical results. However, although these single machine learning and deep learning methods can achieve better forecasting performances than the econometric models, these methods also suffer from the problems of overfitting and hyperparameter optimization.

Deep learning models are considered for processing nonlinear datasets but may prove difficult for completely discerning and extracting characteristics from complex nonlinear data, such as crude oil futures price data. In recent studies, a novel ensemble predicting approach, namely, decomposition ensemble, has been introduced to manage the task of nonlinear and nonstationary time-series data forecasting. The original time-series data are decomposed into several stationary components by the decomposition-ensemble approach through different decomposition technologies, such as the empirical mode decomposition type (EMD-type) (Tang et al., 2015, 2017; Sun S., 2018; Ming et al., 2016; Yu et al., 2008, 2016; Xiong et al., 2013), wavelet analysis (Jiang and Yoon, 2020; Jammazi and Aloui, 2012) and variational mode decomposition (VMD) (Bisoi et al., 2019; Li et al., 2019a,b, 2021). These components can be forecasted by individually predicting models and integrated to produce the forecasting output. For example, Li et al. (2019a,b) decomposed monthly crude oil futures price data into several modes using VMD and applied an SVM optimized by a genetic algorithm (GASVM) and a backpropagation neural network optimized by a genetic algorithm (GABP) to forecast each mode. The empirical results indicate that VMD-based models outperform ensemble empirical mode decomposition (EEMD)-based models and single models through the Diebold–Mariano (DM) test. Sun et al. (2018) applied an interval

ensemble-decomposition method, which integrates BEMD, an interval multilayer perceptron, and an interval exponential smoothing approach (Holt I) to predict the crude oil futures price.

Furthermore, certain underlying factors, such as investor sentiment, speculation activities, and geopolitical tensions, may act as potential causes of changes and fluctuations in oil prices (He and Casey, 2015; Song et al., 2019). These factors are difficult to assess and calculate in empirical works owing to the unquantifiable characteristics of market sentiment and reaction. Currently, abundant available data and natural language processing (NLP) approaches are developed for the structure of investor sentiment indicators. First, the news headline dataset from the energy news website can be marked. Second, using the news headline dataset, daily investor sentiments are scored according to vector space models (Li et al., 2021). Last, the crude oil futures price is predicted by combining itself with the daily polarity score of market investor sentiment. In this paper, we employ the NLP technique to calculate the daily sentiment polarity score of crude oil and use it as a new predictor for the crude oil futures price.

Following the above discussion, a novel decomposition-ensemble approach combined with a sentiment analysis, namely, an EEMD-SOA-GRU-MLR approach, for crude oil futures price forecasting is developed and investigated in this paper. First, the cumulative sentiment score (CSS) sequence is obtained through a sentiment analysis of news headline data. Second, we extract features from the crude oil futures price data and reduce the impact of noise. An adaptive signal decomposition method, namely, EEMD, is applied to decompose the historical crude oil futures price data into intrinsic mode functions. Third, the seagull optimization approach (SOA) is introduced to adjust the hyperparameters of GRU. Subsequently, the optimized GRU can be established to acquire the predicting values of each component integrated with the sentiment polarity score sequence. MLR is then introduced as the ensemble approach that integrates the forecasting results of each component. The suggested contrast hybrid models are compared with the proposed model with different decomposition methods and optimizing approaches, such as the whale optimization algorithm (WOA) and EMD. Both the daily news headline data and daily price data of the West Texas Intermediate (WTI) crude oil future from January 4, 2010, to September 17, 2019, are chosen as examples to illustrate the effectiveness of the proposed decomposition-ensemble model in detail. In addition, the robustness of the proposed model can be demonstrated by verifying whether it can perform well at different horizons. Furthermore, multistep-ahead forecasts can capture the exact dynamic behaviors of crude oil futures prices, which helps investors and government agencies make and amend decisions for different periods. Hence, multistep-ahead daily crude oil futures price forecasting is also considered in this paper. A DM test is employed to prove whether significant differences exist in the forecasting accuracies of the comparison models and the suggested model. Moreover, the WTI futures price dataset is selected to qualitatively analyze the impact of black-swan events on crude oil price fluctuations. The four main contributions of this paper are listed as follows.

- (1) Hybrid models combined with different deep learning models, optimization algorithms, data decomposition technologies, and integration methods are developed for daily crude oil futures price forecasting.
- (2) Hybrid models that use the combination of EMD, SOA, and GRU are demonstrated to be superior to those that use the combination of EEMD, WOA, and LSTM, as well as single deep learning models optimized by varying optimizations in terms of their forecasting accuracies.
- (3) The influencing factor of crude oil price, namely, the sentiment score of the oil market, based on the NLP technique, is investigated and incorporated into the decomposition-ensemble models, thus further improving the prediction accuracy of the daily crude oil futures price.

- (4) The proposed decomposition-ensemble model achieves better forecasts in the test set compared with other comparison models under different forecasting horizons, which demonstrates that the integration method significantly improves the forecasting accuracy.

The remainder of the paper is arranged as follows: Section 2 introduces the relevant methodology formulation in this study. The data sources and model analysis indicators are presented in Section 3. Section 4 illustrates the data and experimental results. Section 5 provides the conclusions of the present study.

2. Methodology

2.1. Research framework

In this section, a novel decomposition-ensemble model, EEMD-SOA-GRU-MLR, is developed for crude oil futures price forecasting. The proposed forecasting model in this paper aims to apply the information extracted from crude oil futures price and news headline datasets to improve the forecasting accuracy of crude oil futures prices. The structure diagram of our proposed model is presented in Fig. 1. The sentiment analysis and EEMD are described in Sections 2.2 and 2.3, respectively. Furthermore, the SOA, GRU, and MLR are introduced in Sections 2.4–2.6.

2.2. Sentiment analysis

Prior to obtaining a sentiment polarity score using sentiment analysis, we preprocess the raw news headlines dataset through three steps: lowercase conversion, the removal of punctuation and stop words. Thus, we convert all the headlines into lowercase using tokenization because lowercase letters are most frequently utilized and are mostly viewed on the same line. Punctuations with stop words, such as “is”, “an” and “the” do not contain key information in the text, and removing these items helps minimize the processing time of the system. After preprocessing the original news headline dataset, the importance of one word to a specific document or one of the documents in a corpus is evaluated. In this paper, an effective weighting technique for information retrieval and data mining, “term frequency-inverse document frequency” (TF-IDF), is applied to assess the significance for this purpose. The importance of one word increases with the number of times that it appears in the headline but decreases with the number of headlines that include the word in the corpus. The TF-IDF value increases with the

occurrence of a word within a headline and decreases with the number of headlines that contain this word in the corpus (Li X. et al., 2019). Specifically, the TF-IDF score of word y in a headline is calculated as follows:

$$TF - IDF = TF * IDF \quad (2.1)$$

where

$$TF = \frac{\text{Frequency of term } y \text{ in a headline}}{\text{Total number of terms in the headline}}$$

$$IDF = \log_e \frac{\text{Total number of news headlines in the corpus}}{\text{Frequency of headlines including term } y \text{ in the corpus}}$$

It is common knowledge that the effect of news related to crude oil on the sentiment of investors is continuous in the futures market, namely, the daily investor sentiment is influenced by the combination of the news on that day and that on the previous several days. Of course, more recent news is more influential than old news. In this study, the CSS is applied to address this problem, following Li Y. et al. (2021). Based on this situation, we assume that all news will have a remarkable impact on investor sentiment for seven days and that its effect will decline each day after its exponential release. Hence, the CSS on day i is the sum of the polarity score on day i and the polarity score of the previous six days:

$$CSS_i = \sum_{j=1}^6 e^{-\frac{j}{7}} SV_{i-j} + SV_i \quad (2.2)$$

where CSS_i is the cumulative sentiment value on day i . $e^{-\frac{j}{7}} SV_{i-j}$ indicates the adjusted polarity score of day $i - j$, and SV_i represents the sentiment score of day i .

In this paper, the *Sentimentr* package in R is applied to calculate the sentiment polarity score of each treated news headline, and it returns the sentiment polarity score in the range of $[-1, 1]$ for each headline. Furthermore, the aforementioned CSS can be obtained by Eq. (2.2) through R, and the CSS is in the range of $[-7, 7]$ in principle. Whether the news is positive or negative depends on the polarity score: the news is considered positive if its polarity scores are greater than 0 and considered negative otherwise. In general, the higher the sentiment polarity score is, the more positive the news is; conversely, the lower the sentiment polarity score is, the more negative the news is.

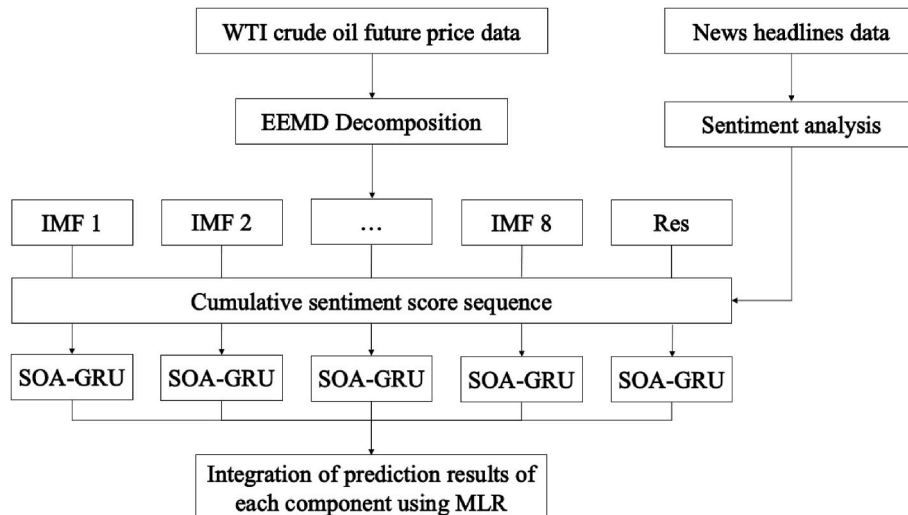


Fig. 1. Structure diagram of the newly proposed forecasting model.

2.3. Ensemble empirical mode decomposition (EEMD)

The EEMD approach is developed based on the EMD method. This approach locally stabilizes the time-series data; subsequently, the time spectrum is obtained by a Hilbert transform to obtain the frequency with a physical meaning. The EEMD approach has self-adaptability because this method decomposes the signal on the basis of the time scale characteristics of the data, and basis functions do not need to be set in advance. Furthermore, an EEMD adds a different series of white noise into the signal to address the defects of mode aliasing and end effects in the EMD method; it performs an ensemble average to eliminate the noise in the overall mean. Therefore, this approach is theoretically suitable for any signal decomposition with various features and has an excellent performance on the decomposition of nonstationary and nonlinear time series.

The steps of the EEMD approach are as follows.

- (1) Initialize the times of white noise N and set $i = 1$ (i is the flag number of the white noise). A batch of white noise series $\omega_i(t)$ is added to the original signal $x(t)$, and the new signal series $X_i(t)$ for $i = 1, 2, \dots, N$ is shown in the following formula:

$$X_i(t) = x(t) + \omega_i(t) \quad (2.3)$$

- (2) Identify the local minima and maxima of the time series $X_i(t)$ and generate the upper and lower envelopes of $X_i(t)$ by a cubic spline interpolation function.
- (3) Compute the mean value $m(t)$ from the upper and lower envelopes and then produce the difference $D(t)$ between $X_i(t)$ and $m(t)$:

$$D(t) = X_i(t) - m(t) \quad (2.4)$$

- (4) If $D(t)$ satisfies the two conditions of the intrinsic mode function, that is, in the entire time range, the number of extreme points and zero-crossing points must be equal or have one difference, and the average of the local maximum and local minimum must be zero at any moment, then $D(t)$ will be the first IMF and replace $X_i(t)$ with $r_i(t) = X_i(t) - \text{IMF}$. If $D(t)$ does not satisfy these conditions, $D(t)$ substitutes for $X_i(t)$.
- (5) Repeat steps (2)–(4) until the residual has only one extreme point, and all the IMF_j ($j = 1, 2, \dots, n$, where n is the number of IMFs) and residual $r(t)$ are obtained.
- (6) Let $i = i + 1$ and repeat (2)–(5) with different white noise until $i = N$.
- (7) Compute the average of IMF and the residual decomposed by N times as the final decomposition results.

$$\overline{\text{IMF}}_j = \frac{1}{N} \sum_{i=1}^N \text{IMF}_{ji} \quad (2.5)$$

$$\overline{r(t)} = \frac{1}{N} \sum_{i=1}^N r_i(t) \quad (2.6)$$

The EEMD approach guarantees the stability of each component and leaves only information about the signal itself. Hence, this method has been widely investigated and considered in the natural and social sciences, such as in crude oil futures price forecasting.

2.4. Seagull optimization algorithm (SOA)

Seagulls are social seabirds. These seabirds migrate among different regions of the world in search of sufficient food with the change in seasons. The predation process of seagulls consists of a migration stage and an attacking stage. In the migration stage, seagulls maintain individual flight independence to avoid collision with each other according

to their natural laws. In the attacking stage, seagulls attack their prey in a spiral flight. Based on the migrating and attacking behavior of seagulls, Dhiman and Kumar invented the SOA, which includes two stages: migration and attacking.

Stage I Migration (Exploration).

This stage simulates how seagulls move from one position to another position. First, during migration, the most important aspect for seagulls is to avoid collisions. To reach this aim, an added variable A is adopted to calculate the new position of the seagull.

$$C_s(t) = A \times P_s(t) \quad (2.7)$$

where $C_s(t)$ indicates a new position that does not collide with another seagull, $P_s(t)$ represents the current position of the seagull, t represents the current iteration, and A indicates the flying behavior of the search seagull in a particular search space.

$$A = c - \left(t \times \left(\frac{c}{\text{Max}_{\text{iteration}}} \right) \right) \quad (2.8)$$

where c is a constant that is employed to control the frequency of variable A , which is linearly decreased from 2 to 0.

Second, in the condition of avoiding a collision with other seagulls, the search seagull moves toward the direction of its best neighbor (i.e., minimal fitness value of other seagulls around it):

$$D_s(t) = B \times (P_{bs}(t) - P_s(t)) \quad (2.9)$$

where $D_s(t)$ indicates the position of search seagull $P_s(t)$ toward the best seagull $P_{bs}(t)$. Variable B is a random value that makes a trade-off between exploration and exploitation. B is calculated as

$$B = 2 \times A^2 \times r_d \quad (2.10)$$

where r_d is a random number in the range of $[0, 1]$.

Third, after the seagull moves in a direction where it will not collide with other seagulls and is close to the fittest seagull, it moves toward and remains close to the direction of the best seagull to reach a new position:

$$N_s(t) = |C_s(t) + D_s(t)| \quad (2.11)$$

where $N_s(t)$ indicates the distance between the search seagull and the best seagull, whose fitness value is less.

Stage II Attacking (exploitation).

The purpose of this stage is to exploit the experience and process of the predation of the search seagull. While attacking their prey, seagulls can persistently change the angle and speed of attack and spiral in the air. The movement behavior in the x , y , and z planes is described as follows:

$$x = r \times \cos(\alpha) \quad (2.12)$$

$$y = r \times \sin(\alpha) \quad (2.13)$$

$$z = r \times \alpha \quad (2.14)$$

$$r = u \times e^{wv} \quad (2.15)$$

where r is the radius of each helix, α is a random angle between 0 and 2π , u and v are the correlation constants of the spiral form, and e is the base of the natural logarithm. The updated position of the seagull is calculated by Eqs. (2.11)–(2.15).

$$P_s(t) = N_s(t) \times x \times y \times z + P_{bs}(t) \quad (2.16)$$

where $P_s(t)$ contains the best solution.

2.5. Gate recurrent unit (GRU)

An RNN has excellent performance in various machine learning

methods in solving time sequence problems. However, the issues of gradient explosion and vanishing are easily addressed during the training process of RNN. As one of the variants of an RNN, the structure of a GRU is more straightforward than that of LSTM, and a GRU can overcome the aforementioned defect. Moreover, a GRU has fewer parameters and higher computational efficiency. Hence, a GRU can reduce the risk of overfitting and is considered in this study to mitigate crude oil futures prices with nonlinear and instability characteristics. Fig. 2 elaborates the internal structure of a GRU.

The cores of the GRU network are reset and update gates. As shown in Fig. 2, the reset gate r_t determines the fusion between the previous memory and the new input of the GRU. The update gate z_t controls the quantity of memory information that is retained. Here, the detailed process of the forward propagation of a single GRU network is introduced as follows:

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (2.17)$$

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (2.18)$$

$$\tilde{h} = \tanh(W_h \cdot [r_t * h_{t-1}, x_t]) \quad (2.19)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h} \quad (2.20)$$

where r_t , z_t , $\tilde{h}$, and h_t represent the reset gate, update gate, current memory status, and output of the unit system, respectively. σ and $\tanh$ are the sigmoid activation function and hyperbolic tangent function, respectively. W_r , W_z , and W_h are the corresponding weight matrices. Moreover, $[\]$ indicates the connection of two vectors, and $*$ represents the multiplication of the corresponding point between two matrices.

2.6. Multiple linear regression (MLR)

In the research of practical problems, the dependent variable is often affected by multiple factors. Thus, it becomes necessary to use two or more factors as independent variables to explain the change in dependent variables, which is known as multiple regression. MLR is the most widely employed traditional forecasting technique for defining the relation between the dependent (response) variables and the independent (explanatory) variables. The relation between the dependent variables and the independent variables is formulated as a multiple regression model as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon \quad (2.21)$$

where Y is the dependent variable; $X_j, j = 1, 2, \dots, k$ are independent variables $\beta_j, j = 0, 1, 2, \dots, k$ are the regression coefficients and ε is the

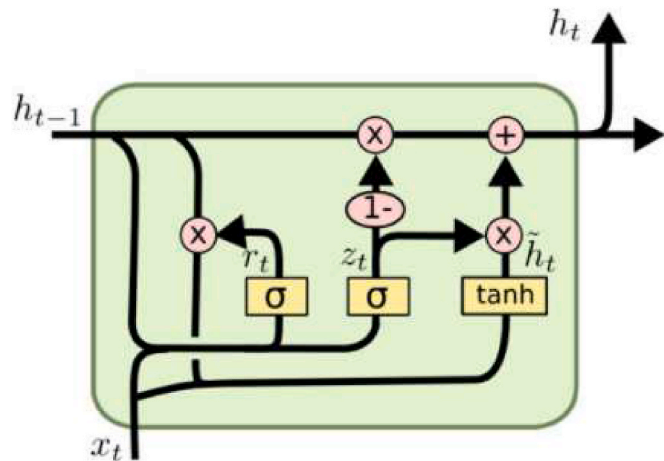


Fig. 2. Internal structure of GRU.

residual. The random error term ε is assumed to be a random variable of the normal distribution with a mean of 0 and variance of σ^2 ; regression coefficients can be obtained by least squares estimation.

In this study, each component of the test set of the proposed model is considered an independent variable, and the test set of the original crude oil futures price data is regarded as the dependent variable. The regression coefficients are obtained by Eq. (2.21). The obtained coefficients are then substituted into the aforementioned equation to acquire the estimated value of the dependent variable, that is, the final forecast results.

2.7. Hybrid forecasting model

The proposed hybrid forecasting includes the following stages.

(1) Sentiment analysis

First, the preprocessor of the news headline data related to crude oil prices should remove stop words and punctuations and calculate the TF-IDF score. Subsequently, based on the assumption that each news item will have a significant impact on investor sentiment for seven days, the CSS sequence is calculated.

(2) Data decomposition

The EEMD method can effectively suppress the mode-mixing phenomenon of EMD. Therefore, the EEMD approach is adopted to decompose the WTI crude oil futures data and to mine the main features of the original price time series.

(3) GRU optimized by the SOA algorithm

Step 1 Initialize the parameters, such as population size and iteration times.

Step 2 Determine the fitness function and set the stop criteria.

$$Fitness = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (2.22)$$

where N is the number of training datasets and $\hat{y}_i$ and y_i are the i th fitting value and observed value, respectively.

Step 3 Update the positions of the seagulls based on the calculated fitness values and determine the stop criteria.

Step 4 If the stop criteria are satisfied, go to the next step. Otherwise, return to Step 3 and continue the next iteration.

Step 5 Stop the process of optimization and reset GRU's parameters using the optimal solutions acquired by SOA. A hybrid model, SOA-GRU, is established.

The hyperparameters of the deep learning model have a significant impact on the forecasting accuracy; thus, the optimization of hyperparameters is crucial. The optimization results of the hyperparameters of the deep learning models are presented in Table 1.

(4) Forecasting

Combined with the calculated CSS, each component decomposed by

Table 1
Hyperparameters of the proposed hybrid deep learning model.

Hyperparameters	Range	Selected
Number of hidden layer neurons	1–100	2
Optimizer	Adam, Adagrad, RMSprop, SGD	Adam
Batch size	16, 32, 64, 128	32
Epoch	100–300	259

the EEMD approach is predicted by the aforementioned SOA-GRU model. The final predicted values can be obtained by integrating the prediction results of each component using MLR.

3. Data and model analysis indicators

3.1. Data sources

In this paper, two datasets are regarded as our raw data: a news headline dataset and the crude oil futures price dataset. Both datasets cover the time span from January 4, 2010, to September 17, 2019, which contains 2548 trading day observations and no missing values. The WTI price dataset of crude oil futures stems from <http://www.eia.doe.gov/>. This dataset is considered an example to test the forecasting abilities of our proposed hybrid model and comparison models. WTI exerts a significant influence on the global economy, and its crude oil prices have fluctuated since 2010, as shown in Fig. 3.

For the news headline dataset, all the news data that were related to crude oil were acquired from oilprice.com, which is one of the world's largest websites for energy news. Instead of using complete news articles in the analysis, we use news headlines given several advantages of the same. There is no missing value for this dataset. First, news headlines

can provide a sufficient summary of the essential news information. Second, news headlines include fewer repetitions and fewer irrelevant words than the news article itself (Li et al., 2021). News headline datasets are selected to constantly measure sentiment. For example, Li et al., 2022 considered news headlines in their text-based prediction framework to forecast agricultural futures prices, and Zhang and Hamori (2021) combined the news headline sentiment index and other factors to analyze the connection effect for economic indicators. Therefore, it is representative to use news headlines to measure sentiment.

In order to definitely show the fluctuation of crude oil future prices and the impact of positive and negative news, the number of positive and negative news for each month are calculated, then compare them to the fluctuation of WTI crude oil futures prices, as shown in Fig. 3. A positive new is defined as a day when the cumulative sentiment score CSS_t is higher than 75% quantile of complete sample of CSS_t . Similarly, a day when CSS_t falls below 25% quantile of complete sample of CSS_t is considered a negative new. The period of rise and fall of oil futures price is basically the same as the number of positive news and negative news shocks are increasing. In conclusion, there is a correlation between news headline sentiment and crude oil futures price trend.

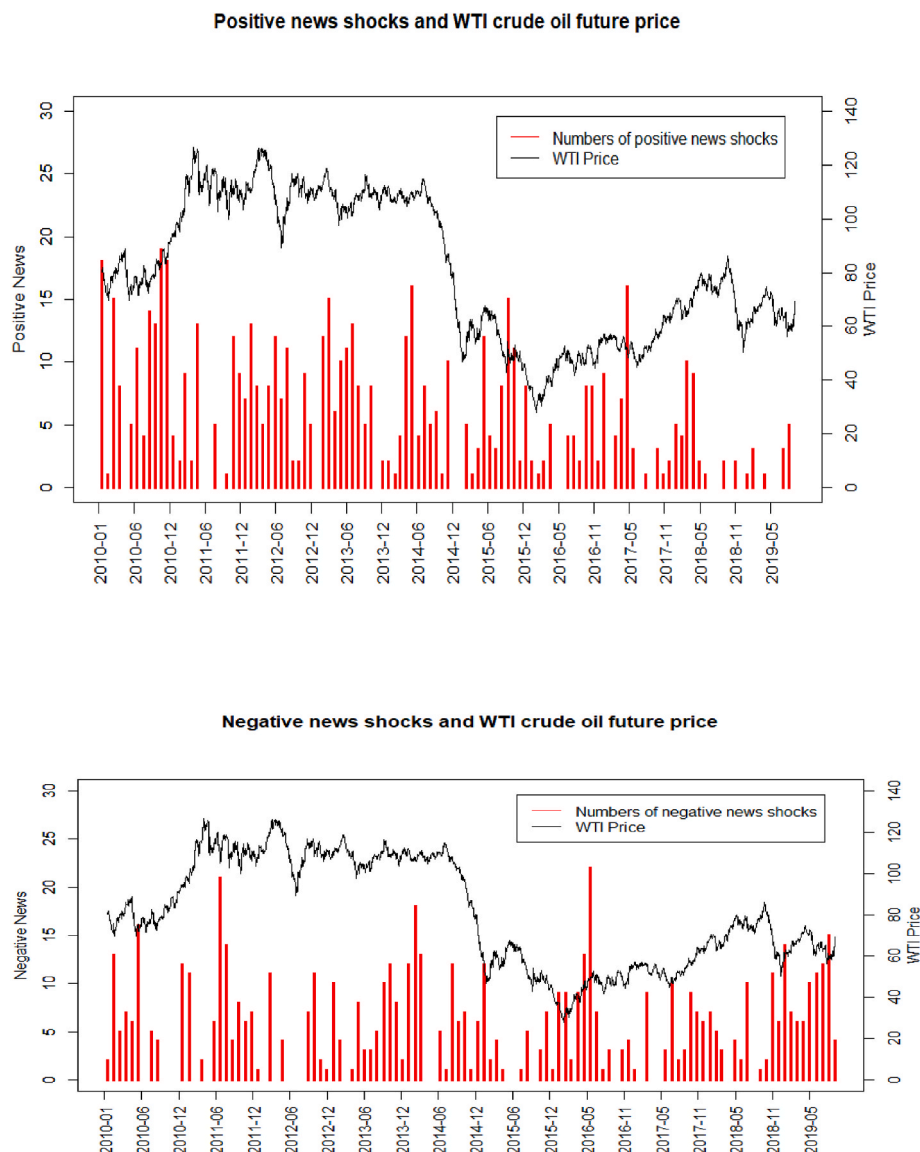


Fig. 3. Daily WTI crude oil future price compared with the monthly number of positive and negative news.

3.2. Model evaluation criteria

Several popular model evaluation criteria, including the root mean square error (RMSE), mean absolute percent error (MAPE), mean absolute error (MAE), and Theil coefficient, are considered to verify the performance of the presented hybrid model.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (3.1)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (3.2)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3.3)$$

$$TH = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}}{\sqrt{\frac{1}{N} \sum_{i=1}^N y_i^2} + \sqrt{\frac{1}{N} \sum_{i=1}^N \hat{y}_i^2}} \quad (3.4)$$

where N is the number of testing datasets, $\hat{y}_i$ is the i th forecasting value, and y_i is the i th observed value. MAE and RMSE measure the absolute error between the predicted value and real value, and the remaining indicators measure the relative error between them. The value ranges of MAE, RMSE, and MAPE are $[0, +\infty]$, respectively, and the value range of the Theil coefficient is $[0, 1]$. The closer their values are to 0, the higher the forecasting accuracy is and the better the forecasting results are.

The Diebold–Mariano (DM) test is performed to test whether there are significant differences in the forecasting ability of different forecasting models to determine the performance of these forecasting models. The null hypothesis indicates that the forecasting ability of the two models is identical, and the alternative hypothesis shows a significant difference in the forecasting ability between the two models. The form of the loss function of the DM test is not strictly limited, and this method can consider various evaluation indicators. Therefore, the test has been widely utilized for the performance comparison of forecasting models.

4. Data description and experimental results

4.1. Data and descriptive statistics

Crude oil futures, as an international financial innovative product produced under violent fluctuations in oil prices, have functions such as price discovery, hedging, and risk aversion. Notably, their financial risks are immeasurable. To better reflect the price fluctuation, returns, and risks of crude oil futures, the return rate sequence r_t and volatility sequence f_t are introduced by processing the crude oil futures price.

$$r_t = \ln p_t - \ln p_{t-1} \quad (4.1)$$

$$f_t = |\ln p_t - \ln p_{t-1}| \quad (4.2)$$

Since 2010, WTI crude oil futures prices have shown an upward trend followed by a downward trend and a distinct volatility clustering effect, as shown in Fig. 3. The descriptive statistical analyses of price, return, and volatility series are shown in Table 2, and the volatility and return series are shown in Fig. 5.

Skewness, kurtosis and Jarque–Bera statistics are familiar descriptive statistics. There is a brief introduction to their theory. Consider the set of samples $x_1, x_2, \dots, x_n$, of which the mean value is μ and the standard deviation is s . Skewness is a characteristic number that describes the deviation of distribution from symmetry; its mathematical expression is shown as follows:

Table 2

Descriptive statistics of the WTI crude oil futures price, return series, and fluctuation series.

	Price	r_t	f_t
Minimum	27.8800	−0.0893	0.0000
Maximum	123.6500	0.1364	0.1364
Mean	80.6046	−0.0001	0.0140
Median	76.9100	0.0000	0.0092
Standard deviation	26.0087	0.0190	0.0135
Skewness	0.0282	0.1120	2.2212
Kurtosis	−1.3977	3.7693	8.0050
Jarque–Bera statistics	207.36	1517.9	8915.7

$$\beta_s = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{s} \right)^3 \quad (4.3)$$

Kurtosis is a characteristic number that describes the distribution steepness or tail thickness; it is defined as follows:

$$\beta_k = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{s} \right)^4 \quad (4.4)$$

Based on the skewness and kurtosis of data samples, the Jarque–Bera test evaluates whether the assumption that the given data obey the normal distribution of unknown mean and variance is tenable. Skewness and kurtosis are subject to a normal distribution, that is, $\beta_s \sim N(0, \frac{6}{N})$ and $\beta_k \sim N(3, \frac{24}{N})$. The kurtosis of the normal distribution is 3, and skewness is 0. *JB* statistics are defined as follows:

$$JB = \frac{S^2}{\frac{6}{N}} + \frac{(K-3)^2}{\frac{24}{N}} \quad (4.5)$$

The *JB* statistics obey a chi-square distribution with 2 degrees of freedom, and the results of the *JB* statistics are not negative.

In Table 2, the mean value of crude oil return is −0.0001; the maximum value is 0.1364, which is greater than the absolute value of the minimum value of 0.0893; and the median is approximately 0. This finding illustrates that the overall opportunities for crude oil futures investment outweigh the risks from January 2010 to September 2019. Nevertheless, the standard deviation is 0.0190, which reflects the high risk of crude oil futures investment. Moreover, the price and return and fluctuation series are left biased and right biased, respectively, which shows that the crude oil futures price violently fluctuates after approximately 2014. The kurtosis of the return and fluctuation series is greater than 3. The Jarque–Bera statistic rejects the assumption that the three series evince a normal distribution at the 1% significance level as they are leptokurtosis, fat-tailed, and skewed.

Figs 3 and 4, and Table 2 shows that the crude oil futures price violently fluctuates, giving rise to black-swan events. For example, crude oil futures prices reached the bottom in the selected period in the second half of 2016. The main reasons are that OPEC, the crude oil supplier, does not reduce production and the output of shale oil in the United States continues to increase, resulting in sufficient crude oil supply, as well as the general economic downturn in Europe, Japan and emerging countries, poor crude oil consumption and oversupply of crude oil in the market. Therefore, because of their unexpected and significant impact, such black-swan events will inevitably cause severe shocks in the financial market, which is an undesirable but inevitable outcome. In the next section, the impact of extreme events caused by black-swan events is further analyzed in terms of the low-frequency component decomposed by EEMD (see Fig. 5).

4.2. Crude oil price decomposition

Crude oil price series are typical nonlinear and unstable data. Therefore, the crude oil futures price can be decomposed into intrinsic

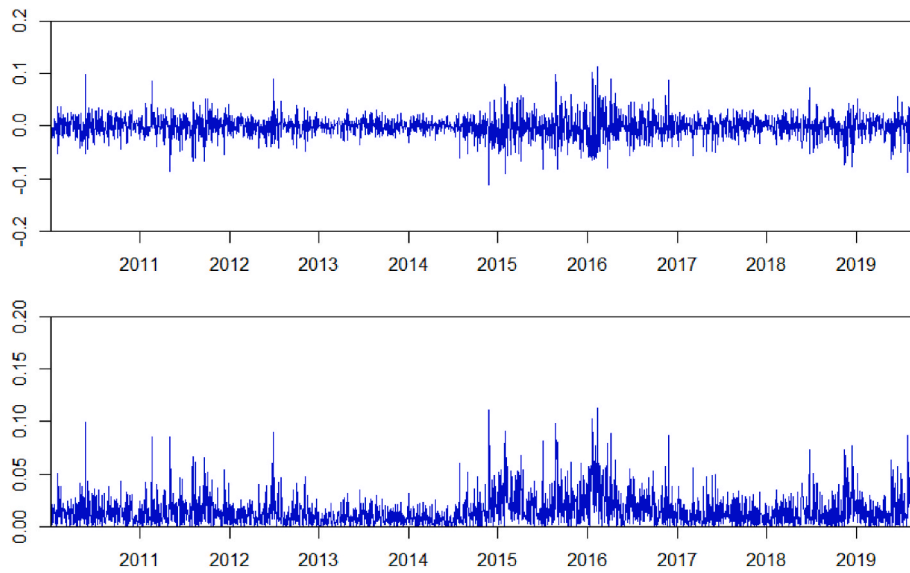


Fig. 4. Time series of returns and volatility. Upper panel: return series. Lower panel: volatility series.

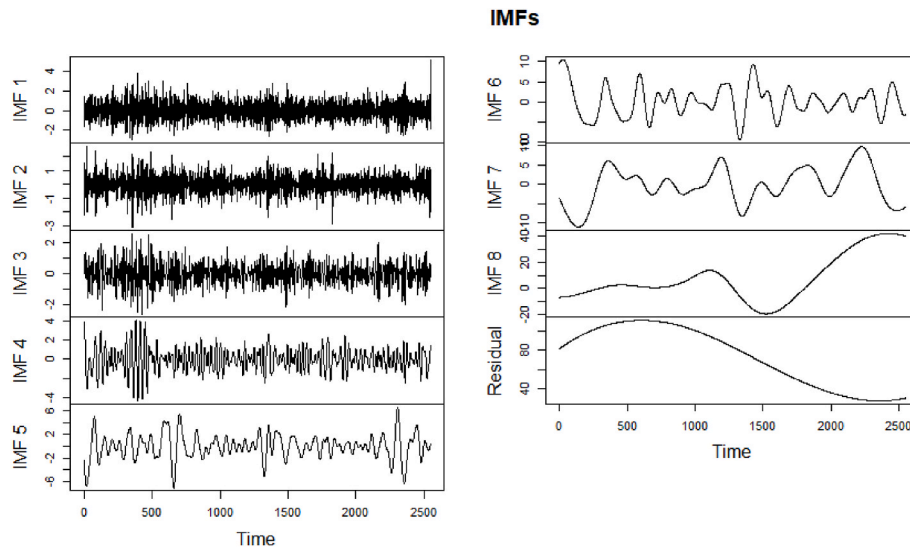


Fig. 5. Decomposition of the intrinsic mode functions of WTI crude oil futures prices from January 4, 2010, to September 17, 2019.

mode functions and residuals with varying frequencies using the EEMD method. The periodicity and nonlinearity of the original data are included in these intrinsic mode functions and residuals. These internal characteristics are obtained by decomposing the daily price data of WTI crude oil futures prices in this study.

Fig. 5. shows the decomposition components of 2548 observations of WTI crude oil futures price data from January 4, 2010, to September 17, 2019. All the intrinsic mode functions can be divided into high and low frequencies, and each component has different characteristics. Each sharp rise or fall of a low-frequency component corresponds to a remarkable event. Therefore, the trend of the low-frequency components corresponds to the impact of major international events and black swan events. The trend of the high-frequency components has a small amplitude, which describes the market's short-selling effect and short-term fluctuations. The residual slowly varies around the long-term mean; thus, it is regarded as a long-term trend of crude oil price evolution and represents the price changes caused by supply and demand in the economic sense (Zhang et al., 2007).

From the perspective of supply and demand, the demand for crude

oil increased at the beginning of 2010 because of the steady growth of the world economy financial crisis of 2008. The demand grew steadily before 2013, and the fluctuation of crude oil prices before 2013 was considered the third oil crisis and the economic crisis of 2008. The demand for crude oil sharply decreased after 2013 because the economic recovery of most countries was lower than expected, causing a decline in demand for commodities. The increase in shale oil production in the United States and the reproduction of crude oil in traditional oil-producing countries, such as Libya, in 2014 caused an imbalance between supply and demand.

To further analyze the impact of this black-swan event and the relation between supply and demand on crude oil futures prices, we reconstruct the signals into high-frequency, low-frequency, and long-trend components, as shown in Fig. 6. The high-frequency component, which is mainly caused by the short-selling effect and short-term fluctuation of the market, generates a fast oscillation near 0 with a small amplitude, which has a short-term impact on the crude oil price. However, the influencing factors of crude oil are extraordinarily complex. Factors such as bad weather, strikes, and inventory depletion can also be



Fig. 6. Reconstruction of crude oil futures price components from January 4, 2010, to September 17, 2019.

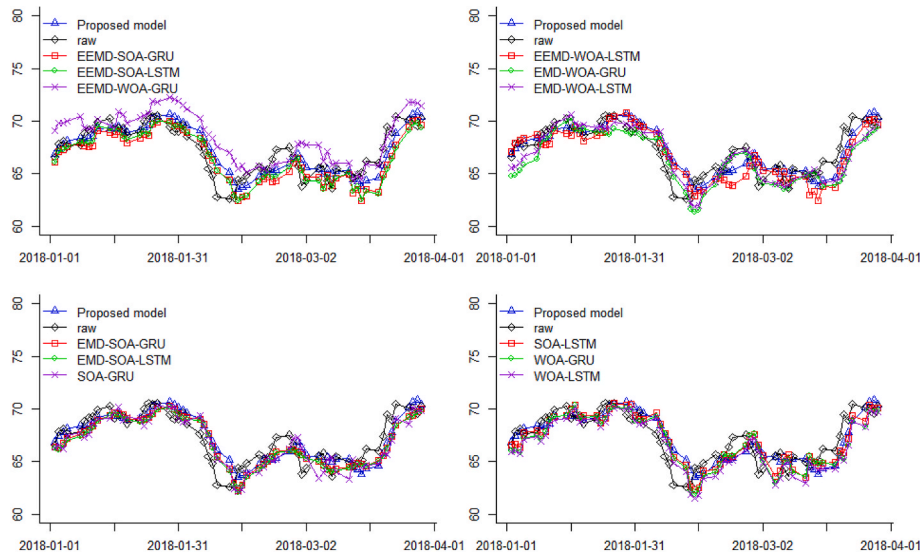


Fig. 7. One-step-ahead predicting values of the proposed and comparison models combined with sentiment analysis.

divided into high-frequency components because their impact on the crude oil price is similarly short. The residual term reflects the long-term evolution trend of oil prices. This upward trend, followed by a downward trend, is consistent with world economic development, implying that the long-term crude oil price trend may be determined by global economic growth. Here, we pay close attention to the low-frequency component. Every sharp rise or fall represents the impact of extreme events that are difficult to quickly eliminate and have a long cycle. In Fig .5, the amplitude of certain data points exceeds \$10 or even \$20, indicating that certain extreme events significantly impact the oil price. The impact of extreme events is separated from the price series in the form of a low-frequency component, which excludes the influence of other frequency terms, making the trend of its curve more consistent with the trend of the original price data curve. Therefore, the impact of subsequent similar extreme events can be observed and even predicted.

The steady global economic growth after the financial crisis in 2008 was achieved, and crude oil prices began to rise, given the opening of the first round of quantitative easing monetary policy in the United States and the enormous energy demand induced by the vigorous development

of China's economy. Many black-swan events that affected crude oil futures prices emerged in an endless stream. For example, since 2010, oil prices have sharply fluctuated over just one or two years. The fluctuations were accompanied by favorable economic expectations, such as the Libyan crisis, killing of Osama bin Laden, deterioration of the European debt crisis, continued unrest in Syria, and Iranian nuclear issue. From 2014 to 2016, crude oil prices continued to plummet, affected by shale oil, and the oil price began to hit the bottom and rebound with the implementation of the production reduction agreement at the end of 2016. However, the escalation of trade disputes between China and the United States in 2018 once again caused a sharp drop in crude oil prices. These events have caused the crude oil futures price sharply fluctuate for a long time. In Fig .6, the sharp upward and downward trend of the crude oil futures price series is consistent with the low-frequency component, which also shows that this trend of the low-frequency component corresponds to the effect of black-swan events. In addition, the crude oil price trend is consistent with the residual component in the long term, which is determined by the global economic trend and the overall supply and demand for quotas.

4.3. Forecasting results and model comparison

4.3.1. Comparison of evaluation indicators

To test the performance of the proposed model, 12 models are considered herein for the sake of comparison: SOA-LSTM, SOA-GRU, WOA-LSTM, WOA-GRU, EMD-WOA-LSTM, EMD-WOA-GRU, EMD-SOA-LSTM, EMD-SOA-GRU, EEMD-WOA-LSTM, EEMD-WOA-GRU, EEMD-SOA-LSTM, and EEMD-SOA-GRU. For notation simplicity, we use model1-model 12 to denote them.

In this section, both datasets are sampled from January 4, 2010, to September 17, 2019. First, we use the *Sentimentr* package in R to obtain the sentiment polarity score from the news headline dataset and decompose the crude oil futures price into several intrinsic mode functions through the EEMD approach. Second, the datasets are divided into a training set and test set, with division proportions of 80% and 20%, respectively. In addition, empirical studies on multistep-ahead daily crude oil futures price forecasting with various forecasting horizons (in our study, $H = \{1, 2, 3\}$) are investigated for the WTI crude oil market. Last, we use the selected model to predict each component with the calculated CSS and integrate the component forecasting results by different integration approaches. The detailed experimental results and analysis are presented as follows.

The related values of the four model evaluation criteria of one-step-ahead forecasts for the predicting models are shown in Table 3; values marked in bold represent the best value of each evaluation metric. Furthermore, to more clearly exhibit the forecasting results of the prediction models and original WTI crude oil futures prices more clearly, Fig. 7 shows the forecasting results for the period from January 2, 2018, to March 31, 2018. However, the forecasting results of the selected models overlap because the forecasting errors of the selected models do not have enormous differences. Similarly, Table 4 and Table 5 present the detailed calculation results of the two-step-ahead and three-step-ahead WTI crude oil prices through different forecasting models. From these tables and figures, some conclusions can be reached.

First, the forecasting performance of the models using the EEMD method is inferior compared with the models using the EMD method and the models without a decomposition approach in terms of the four model criteria in all cases, as shown in Tables 3–5. For example, the RMSE value of the models using the EEMD method under different horizons is higher than those using the EMD method and those without the decomposition approach in the WTI crude oil market. The main reason for this situation might be that although the EEMD method can add different series of white noise into the signal to address the defect of mode aliasing, the white noise series combined with the sentiment scores of investors may decrease the forecasting accuracy. Furthermore, the forecasting performance of the models that applied the EMD method is superior to those that did not use a decomposition approach. These results strongly illustrate that models that applied the EMD approach have a better forecasting performance than those that applied the EEMD approach and those that did not apply a decomposition approach.

Second, the forecasting performance of the models that employed the

SOA algorithm is superior to those that employed the WOA algorithm in terms of the four model criteria in all cases, as shown in Tables 3–5. For example, the MAE values of the EMD-SOA-GRU under different horizons are lower than those of the EMD-WOA-GRU, and the Theil values of the EEMD-SOA-LSTM under different horizons are lower than those of the EEMD-WOA-LSTM. These results confirm that the intelligent optimization algorithm SOA has a better optimizing performance than the previously popular WOA.

Third, in general, the forecasting performance of the models that applied the GRU approach is superior to those that applied the LSTM approach regarding the four model criteria in all cases, as shown in Tables 3–5. For example, the MAPE values of the EEMD-SOA-GRU under different horizons are lower than those of the EEMD-SOA-LSTM. The MAPE values of the SOA-GRU are higher than those of the SOA-LSTM in one-step-ahead and two-step-ahead forecasting, but the difference between the values of the two models is minimal. The main reason might be that the GRU approach has a simpler structure and fewer parameters than the LSTM. However, no model can be superior to another model in any case of any problem. Therefore, these results prove that the deep learning approach GRU achieves a better forecasting performance than LSTM in most cases.

Fourth, the forecasting performance of the comparison models is inferior to that of the proposed model in terms of the four model criteria in all situations, as shown in Tables 3–5. This result not only confirms the effectiveness of the suggested method for crude oil price forecasting but also proves that the integration method MLR can improve the forecasting accuracy and significantly reduce the forecasting error. In summary, we demonstrate that EMD, SOA, and GRU achieve higher forecasting accuracies than EEMD, WOA, and LSTM in the majority of cases and that MLR can further improve the forecasting performance compared with the linear combination of each mode.

To validate the effectiveness of sentiment in improving forecasting accuracy, the empirical forecasting results without CSS under different horizons are presented as follows. Notably, the empirical forecasting results with CSS are superior to the results without CSS under different horizons, with the exception of EEMD-type models, as shown in Tables 3–8. This situation may be attributed to the notion that CSS may amplify the influence of the random white noise sequence added in the EEMD method, resulting in a decline in forecasting accuracy. Similarly, we can conclude that the SOA has a better optimizing performance than the WOA and that the proposed model has the best forecasting accuracy in terms of the four model criteria in all cases, as shown in Tables 6–8. However, the forecasting performance of the models that use the EEMD method is superior to that of the models that use the EMD method. The reason may be that EEMD can effectively solve the problem of mode aliasing when addressing nonlinear and unstable signals without other independent variables. In addition, there are no significant differences between LSTM and GRU in forecasting accuracy.

In conclusion, first, regardless of whether CSS is used to predict the price of crude oil futures, we can draw two conclusions: SOA is superior to WOA in improving forecasting accuracy, and MLR can effectively

Table 3
One-step-ahead forecasting performance of different models.

Model	SOA-LSTM (Model 1)	SOA-GRU (Model 2)	WOA-LSTM (Model 3)	WOA-GRU (Model 4)	EMD-WOA-LSTM (Model 5)	EMD-WOA-GRU (Model 6)	EMD-SOA-LSTM (Model 7)
RMSE	1.2783	1.2926	1.3924	1.3058	1.2345	1.1786	1.0491
MAE	0.8965	0.9422	1.0360	0.9432	0.9952	0.9284	0.8290
MAPE	0.0135	0.0142	0.0156	0.0142	0.0149	0.0139	0.0124
Theil	0.0094	0.0095	0.010	0.0096	0.0091	0.0086	0.0077
Model	EMD-SOA-GRU (Model 8)	EEMD-WOA-LSTM (Model 9)	EEMD-WOA-GRU (Model 10)	EEMD-SOA-LSTM (Model 11)	EEMD-SOA-GRU (Model 12)	Proposed	
RMSE	1.0045	2.4434	1.5222	2.2392	1.8546	1.0036	
MAE	0.7766	2.0785	1.1981	1.8926	1.5458	0.7546	
MAPE	0.0116	0.0307	0.0181	0.0279	0.0228	0.0113	
Theil	0.0074	0.0181	0.0111	0.0166	0.0137	0.0074	

Table 4

Two-step-ahead forecasting performance of different models.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
RMSE	1.6911	1.7753	1.9925	1.7830	1.5595	1.4707	1.3576
MAE	1.2883	1.4038	1.6694	1.3326	1.2575	1.1757	1.0514
MAPE	0.0193	0.0210	0.0247	0.0200	0.0187	0.0175	0.0157
Theil	0.0124	0.0130	0.0147	0.0130	0.0115	0.0108	0.0100
Model	Model 8	Model 9	Model 10	Model 11	Model 12	Proposed	
RMSE	1.3201	4.1816	4.7013	2.2842	1.4240	1.2893	
MAE	1.0036	3.6711	4.4610	2.4631	1.1241	0.9544	
MAPE	0.0150	0.0537	0.0669	0.0361	0.0167	0.0143	
Theil	0.0097	0.0314	0.0333	0.0212	0.0105	0.0094	

Table 5

Three-step-ahead forecasting performance of different models.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
RMSE	2.1000	2.1315	2.1697	2.0692	2.0978	1.7720	1.6078
MAE	1.6748	1.6025	1.6643	1.6276	1.7309	1.3766	1.2551
MAPE	0.0251	0.0241	0.0249	0.0244	0.0256	0.0205	0.0188
Theil	0.0154	0.0155	0.0158	0.0152	0.0155	0.0130	0.0118
Model	Model 8	Model 9	Model 10	Model 11	Model 12	Proposed	
RMSE	1.6260	5.2645	5.8085	2.6948	1.9054	1.5342	
MAE	1.2840	4.5960	5.2047	2.2345	1.5617	1.1653	
MAPE	0.0192	0.0676	0.0764	0.0328	0.0231	0.0175	
Theil	0.0119	0.0398	0.0442	0.0200	0.0141	0.0112	

Table 6

One-step-ahead forecasting performance of different models without CSS.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
RMSE	2.1702	2.2333	2.1953	2.2243	1.8600	1.9754	1.8241
MAE	0.9535	0.9676	1.0645	1.0036	1.0443	1.1329	0.8948
MAPE	0.0237	0.0243	0.0255	0.0248	0.0258	0.0282	0.0234
Theil	0.0203	0.0209	0.0206	0.0209	0.0175	0.0185	0.0171
Model	Model 8	Model 9	Model 10	Model 11	Model 12	Proposed	
RMSE	1.8496	1.7773	1.7433	1.6195	1.5814	1.5133	
MAE	0.9134	1.0476	0.9837	0.9559	0.8836	0.8287	
MAPE	0.0240	0.0250	0.0238	0.0231	0.0215	0.0199	
Theil	0.0174	0.0166	0.0163	0.0151	0.0148	0.0142	

Table 7

Two-step-ahead forecasting performance of different models without CSS.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
RMSE	2.1381	2.1884	2.2924	2.2726	2.0662	1.8437	1.8589
MAE	1.0473	0.9595	1.0833	1.1310	1.3374	1.0591	0.7893
MAPE	0.0251	0.0240	0.0264	0.0267	0.0301	0.0266	0.0224
Theil	0.0201	0.0205	0.0216	0.0213	0.0195	0.0173	0.0174
Model	Model 8	Model 9	Model 10	Model 11	Model 12	Proposed	
RMSE	1.8157	1.8568	1.8415	1.8577	1.9457	1.7837	
MAE	0.7912	1.1583	1.0640	0.9401	0.9745	0.9004	
MAPE	0.0223	0.0267	0.0248	0.0221	0.0234	0.0219	
Theil	0.0173	0.0173	0.0172	0.0173	0.0182	0.0167	

Table 8

Three-step-ahead forecasting performance of different models without CSS.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
RMSE	2.2312	2.3377	2.4338	2.2673	2.1433	1.9437	2.0469
MAE	1.1029	1.0334	1.3865	1.1309	1.3890	0.9942	0.9514
MAPE	0.0266	0.0264	0.0325	0.0272	0.0311	0.0257	0.0254
Theil	0.0210	0.0219	0.0228	0.0213	0.0220	0.0183	0.0192
Model	Model 8	Model 9	Model 10	Model 11	Model 12	Proposed	
RMSE	1.9223	2.0968	2.2326	1.9587	1.9853	1.8446	
MAE	0.9689	1.2244	1.2566	0.9980	1.0064	0.9512	
MAPE	0.0248	0.0288	0.0295	0.0242	0.0243	0.0226	
Theil	0.0180	0.0196	0.0208	0.0183	0.0186	0.0172	

achieve better forecasting performance. Second, EMD and EEMD show opposite prediction effects in the two cases, which may be caused by the random white noise sequence added to the EEMD method. White noise without CSS can address the defect of mode aliasing, but combining white noise and CSS may amplify the influence of random white noise and cause a further decline in forecasting accuracy. In addition, LSTM and GRU have different forecasting accuracies in the two cases. According to the no free lunch (NFL) theorem, compared with LSTM, the simpler structure of the GRU has no significant advantage in crude oil futures price forecasting without other independent variables. However, with an increase in the number of independent variables, the GRU may have a better forecasting performance. Last, we can validate the effectiveness of sentiment in improving forecasting accuracy, and the proposed model has the best forecasting performance under different horizons, as shown in Tables 3–8

4.3.2. Analysis of the DM test

Furthermore, the DM test is performed to test the superiority of the proposed model and comparison models from a statistical perspective, and the corresponding results on the basis of MAPE are presented in Tables 9–11 (Sun et al., 2022). The aforementioned four main conclusions can be statistically confirmed through the DM test.

First, the DM statistical values of the Model 12 and Model 8 are all higher than 2.58 with respect to the MAPE values under different horizons. The p values are smaller than the significance of 1%, indicating that the models that apply EMD can be better verified in statistical terms than those that apply EEMD in crude oil price forecasting with a confidence level of 99%. Similarly, the DM statistical values of the Model 8 and Model 2 are higher than 2.58, indicating that the models that use EMD can be better statistically verified than single models. Second, the DM statistical values of the Model 6 and Model 8 are almost higher than 2.58 under different horizons, suggesting that the optimization performance of the SOA has been better statistically verified than that of the WOA. Third, the DM statistical values of the Model 12 and Model 11 are all higher than 2.58 in relation to the MAPE values under different horizons, demonstrating that the deep learning approach GRU has been better statistically verified for its forecasting performance compared with LSTM in general. Fourth, from the results of the DM test, we can conclude that when the proposed model is treated as the target model while the selected comparison models are treated as benchmark models,

almost all the DM statistical values are much higher than 2.58 in all cases. These findings demonstrate that the proposed model is superior to all comparison models at the 90% confidence level in multi-step ahead forecasting and outperforms almost benchmark models except EMD-SOA-GRU at the 99% confidence level in one-step forecasting. The DM statistical values of EMD-SOA-GRU and the proposed model are lower than 1.69 for one-step-ahead crude oil price forecasting. However, the performance of the proposed model is slightly better than that of the other two models. These results demonstrate that the newly proposed model is a promising tool for crude oil futures price forecasting.

5. Conclusions

As one of the most essential commodities, the price fluctuation of crude oil not only pushes inflation but also has a significant impact on the politics and economies worldwide. Because crude oil prices are affected by many factors, a decomposition-ensemble model named EEMD-SOA-GRU-MLR is proposed for forecasting crude oil futures prices that have nonlinearity, instability, and randomness. An international representative daily WTI crude oil futures price dataset and a news headline dataset related to crude oil prices are selected as the research objects for positive analysis in this study. Moreover, return and volatility are introduced to analyze the fluctuation of crude oil futures prices and the investment values of crude oil futures. The components decomposed by EEMD are reconstructed according to high and low frequencies to analyze the impact of black-swan events on the crude oil price and its long-term trend. In addition, a comparison of the forecasting of whether CSS is considered in different horizons validates the effectiveness of sentiment in improving forecasting accuracy. The DM test is conducted with different horizons to ascertain whether there are differences among models in terms of forecasting accuracy. Based on the empirical results, we can draw the following conclusions:

- (1) The proposed hybrid forecasting model, EEMD-SOA-GRU-MLR, has a higher forecasting accuracy and lower forecasting error than the selected comparison models in all cases.
- (2) From January 4, 2010, to September 17, 2019, crude oil futures have had a high investment income and substantial investment risk.

Table 9

DM test for different models in one-step-ahead forecasting for WTI crude oil futures prices.

	Model12	Model11	Model10	Model9	Model8	Model7	Model6	Model5	Model4	Model3	Model 2	Model 1
Proposed	15.815 (***)	19.377 (***)	11.649 (***)	21.407 (***)	0.976	2.765 (***)	5.317 (***)	6.834 (***)	5.885 (***)	8.175 (***)	6.104 (***)	4.640 (***)
Model 12		20.069 (***)	5.0006 (***)	22.773 (***)	16.562 (***)	16.111 (***)	10.926 (***)	11.821 (***)	10.127 (***)	8.742 (***)	10.670 (***)	10.860 (***)
Model 11			9.3035 (***)	9.1658 (***)	20.467 (***)	20.171 (***)	15.297 (***)	17.344 (***)	14.326 (***)	13.214 (***)	14.871 (***)	14.909 (***)
Model 10				11.892 (***)	9.552 (***)	8.287 (***)	5.492 (***)	4.054 (***)	4.903 (***)	3.103 (***)	5.042 (***)	5.849 (***)
Model 9					22.269 (***)	21.955 (***)	21.955 (***)	18.375 (***)	16.110 (***)	15.117 (***)	16.838 (***)	16.840 (***)
Model 8						8.929 (***)	6.417 (***)	9.762 (***)	5.033 (***)	8.121 (***)	5.434 (***)	3.730 (***)
Model 7							4.105 (***)	7.700 (***)	3.337 (***)	6.459 (***)	3.656 (***)	2.085 (***)
Model 6								2.875 (***)	2.039 (**)	3.040 (***)	2.019 (**)	2.744 (***)
Model 5									1.763 (*)	1.822 (*)	1.775 (*)	2.290 (**)
Model 4										4.460 (***)	0.517	3.243 (***)
Model 3											7.633 (**)	7.903 (***)
Model 2												4.533 (***)

***1% significance level, **5% significance level, *10% significance level.

Table 10

DM test for different models in two-step-ahead forecasting for WTI crude oil futures prices.

	Model12	Model11	Model10	Model 9	Model 8	Model 7	Model 6	Model 5	Model 4	Model 3	Model 2	Model 1
Proposed	6.657 (***)	20.170 (***)	49.634 (***)	27.523 (***)	1.927 (*)	3.398 (***)	6.383 (***)	7.907 (***)	8.655 (***)	14.232 (***)	10.358 (***)	8.160 (***)
Model 12		22.973 (***)	40.449 (***)	30.407 (***)	4.008 (***)	2.345 (**)	2.043 (**)	4.043 (***)	4.280 (***)	12.241 (***)	6.428 (***)	3.764 (***)
Model 11			18.014 (***)	27.531 (***)	20.667 (***)	20.766 (***)	21.760 (***)	18.627 (***)	12.669 (***)	12.376 (***)	14.013 (***)	14.352 (***)
Model 10				6.826 (***)	44.709 (***)	42.166 (***)	37.881 (***)	35.896 (***)	42.702 (***)	29.660 (***)	36.412 (***)	41.532 (***)
Model 9					27.675 (***)	27.652 (***)	28.693 (***)	26.417 (***)	21.465 (***)	22.004 (***)	22.927 (***)	22.905 (***)
Model 8						6.549 (***)	9.002 (***)	10.689 (***)	7.956 (***)	15.931 (***)	11.072 (***)	7.880 (***)
Model 7							8.190 (***)	10.853 (***)	6.395 (***)	15.452 (***)	9.632 (***)	-6.222 (***)
Model 6								4.762 (***)	3.286 (***)	13.249 (***)	6.003 (***)	2.756 (***)
Model 5									1.678 (*)	10.687 (***)	3.778 (***)	1.851 (*)
Model 4										5.974 (***)	2.107 (**)	2.346 (***)
Model 3											8.998 (***)	9.064 (***)
Model 2												6.595 (***)

***1% significance level, **5% significance level, *10% significance level.

Table 11

DM test for different models in three-step-ahead forecasting for WTI crude oil futures prices.

	Model12	Model11	Model10	Model 9	Model 8	Model 7	Model 6	Model 5	Model 4	Model 3	Model 2	Model 1
Proposed	9.000 (***)	14.112 (***)	31.913 (***)	27.561 (***)	3.896 (***)	3.163 (***)	5.751 (***)	10.542 (***)	9.876 (***)	9.853 (***)	8.641 (***)	10.705 (***)
Model 12		17.846 (***)	36.992 (***)	31.700 (***)	6.740 (***)	6.734 (***)	3.480 (***)	4.621 (***)	1.713 (*)	1.772 (*)	2.966 (***)	3.393 (***)
Model 11			28.445 (***)	24.532 (***)	12.415 (***)	11.820 (***)	9.654 (***)	10.271 (***)	5.866 (***)	4.135 (***)	4.097 (***)	5.846 (***)
Model 10				11.396 (***)	25.312 (***)	25.015 (***)	24.136 (***)	25.537 (***)	23.202 (***)	21.809 (***)	21.466 (***)	23.400 (***)
Model 9					21.729 (***)	21.463 (***)	20.764 (***)	22.022 (***)	19.608 (***)	18.359 (***)	18.019 (***)	19.801 (***)
Model 8						1.323 (***)	4.994 (***)	10.248 (***)	7.868 (***)	7.656 (***)	6.725 (***)	8.666 (***)
Model 7							5.675 (***)	9.140 (***)	8.310 (***)	8.191 (***)	7.270 (***)	8.972 (***)
Model 6								5.820 (***)	5.119 (***)	6.171 (***)	5.163 (***)	5.773 (***)
Model 5									0.321 (**)	1.875 (**)	1.053 (**)	1.810 (**)
Model 4										3.429 (***)	1.991 (**)	3.006 (***)
Model 3											1.273 (**)	2.067(**)
Model 2												1.033 (***)

***1% significance level, **5% significance level, *10% significance level.

- (3) The sharp rise and fall of the low-frequency component of EEMD correspond to black-swan events.
- (4) Different deep learning models, optimization algorithms, and data decomposition technologies have a significant effect on forecasting results through the DM test, and the selected integration method, namely, MLR, significantly improves the forecasting accuracy in most cases.

The empirical results demonstrate that the new decomposition-ensemble model suggested in this study has outstanding prediction performance. The results of this study may provide a reference for the energy planning of a country or the crude oil purchase plan of related enterprises. In addition, the proposed decomposition-ensemble approach can be utilized for other complex time series forecasting problems, such as electricity price forecasting and bitcoin price

forecasting, which may achieve better performance.

Despite the capability of the suggested decomposition-ensemble approach, there is still room for improvement. It is universally known that irregular factors, such as inventories, poor climate, and international politics, affect oil price fluctuations; however, it is difficult to quantify their effect. Consequently, only considering two variables to forecast crude oil futures prices is one of the limitations of this paper. Another limitation of this study is that the point forecasting of crude oil prices may disregard the fluctuation information. Therefore, we suggest that a higher level of predictive accuracy can be obtained by quantifying these factors and by considering interval or probability forecasting.

Author statement

He Jiang: Conceptualization; Data curation; Formal analysis;

Funding acquisition; Investigation; Methodology; **Weiqiang Hu**: Project administration; Resources; Software; Supervision; Validation; Visualization; **Ling Xiao**: Software; Supervision; Validation; Visualization; **Yao Dong**: Roles/Writing - original draft; Writing - review & editing.

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